FDE ACADEMY · DELIVERY OFFICE · ENGAGEMENT PACK

Sahyog Care Companion

Issued to the deployed FDE team: client context, mandate, data estate, target architecture, delivery plan and acceptance for a two-week agentic build on LangGraph.

ENGAGEMENT AT A GLANCE

- Client: Sahyog Health Network, 6 hospitals and 22 clinics, fictional
- Duration: two weeks from kickoff to steering review
- Team: pods of three, the FDE lead rotating weekly
- Stack: LangGraph 1.2, Postgres with pgvector, FastAPI, MCP
- Acceptance: steering review plus a five-minute technical defence

Healthcare · patient and doctor workflows

RAG · tools · human-in-the-loop · MCP

Three depth tiers: Core, Pro, Architect

Library versions and datasets verified against PyPI, Kaggle and OpenML on 21 October 2026. The client, its people and its figures are fictional.

“If I'd asked customers what they wanted, they would have said ‘a faster horse’”

— Henry Ford

“The customer will always tell you what they want. Your job as FDE is to find what they need.”

Go by requirements as what Client WANT here and figure it out what Client needs here actually.

CONTEXT

Five kinds of page, and the map of the pack

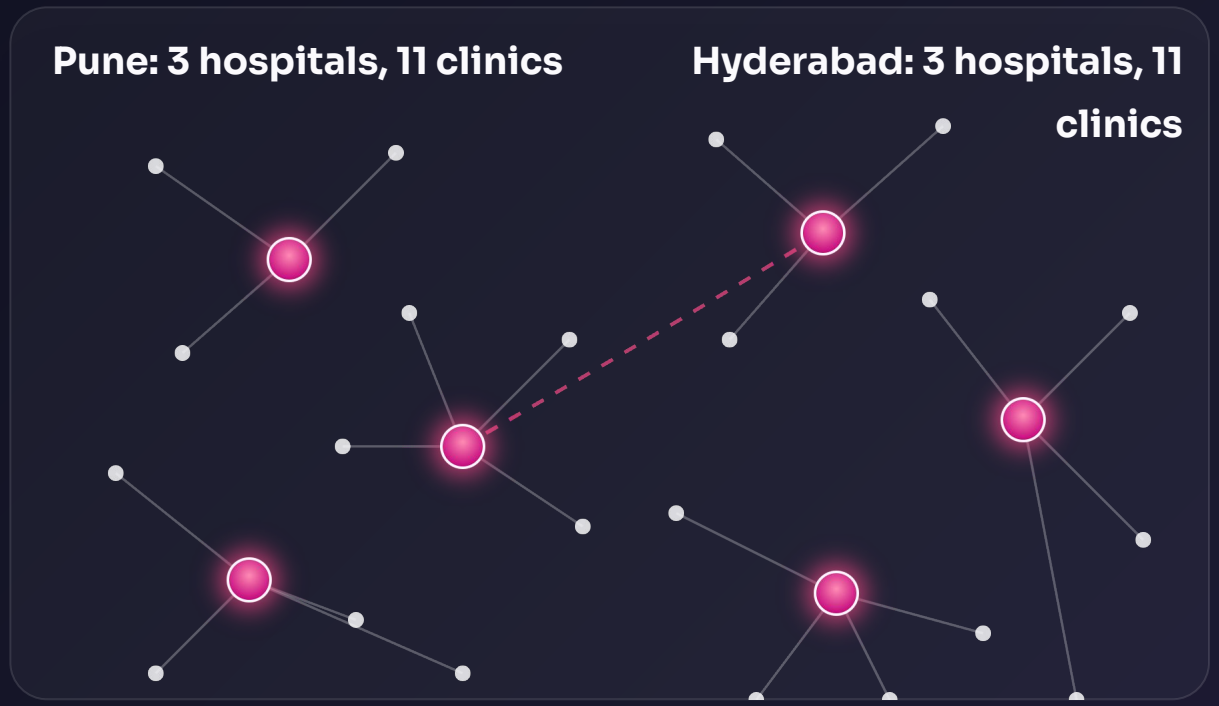
CONTEXT	The client's material and the facts the engagement is built on; read once, return when a decision needs them.
DELIVERABLE	A discovery artefact, D1 to D7, that the team files before design starts and that the steering review scores.
DESIGN	The target architecture the team implements, then corrects and re-issues in its own architecture pack.
ASSURANCE	How the work is tested, by the team during the build and at acceptance afterwards.
REVIEW	The questions the client's architect and the examiner put to the team, with the depth a senior answer carries.

Section	Pages
01 How to use this pack	1 to 2
02 Client context	3 to 7
03 Discovery deliverables D1 to D3	8 to 10
04 Requirements	11 to 16
05 Data estate	17 to 22
06 Solution architecture	23 to 31
07 Platform and risk	32 to 36
08 Ways of working	37 to 39
09 Delivery plan	40 to 44
10 Assurance	45 to 47
11 Acceptance and review	48 to 51
12 Roadmap and annex	52 to 56

First pass: about ninety minutes, cover to close. After that the pack is a reference: deliverable pages during discovery, design pages during the build, assurance pages before the review.

CONTEXT

The client: a two-city network whose front desk has stopped scaling



Large glowing nodes are hospitals, small dots are clinics, and the dashed line is the one shared platform the client wants across both cities.

3.1M
outpatient visits a year

9,000
appointments a day

41%
of bookings still made by phone

18%
of booked patients never arrive

2.4M
report PDFs scanned a year

11 min
of each consultation spent typing

The board's mandate to the programme is in the three pink tiles: take the phone out of booking, bring patients to their own records, and return the eleven minutes to the doctor. Every figure is the client's own and is verified in discovery.

The mandate: five outcomes the client is buying, and the lines around them

Outcomes

1. A patient finds the right doctor and books in under three minutes without a phone call.
2. A patient sees and questions their own records, every answer cited and scoped to their own data.
3. A doctor leaves each consultation with a signed note after at most ninety seconds of review.
4. Every AI output is traceable, auditable and reversible by a named human.
5. The design runs under the DPDP Act and is ABDM-ready, with a path to on-premise handling of identified data.

In scope

- Find-my-doctor triage and slot booking.
- Record upload, storage and questions over the record.
- Consultation scribe with doctor sign-off.
- Shared platform: identity, consent, audit, evaluation.

Out of scope

- Billing and insurance adjudication.
- Telemedicine video and wearable data.
- Inpatient workflows, offered later as extensions only.

Constraints

- Identified health data never leaves India.
- The LLM vendor must be swappable without a rewrite.
- The hospital system exposes FHIR R4 read-only APIs for demographics and encounters.
- Pharmacy stock arrives as a nightly CSV today; an API is promised later.
- English first, Hindi and Marathi later, so nothing in the schema may assume English.
- Peak load is 1,400 requests an hour in the morning window.

CONTEXT

The RFP addendum: seven answers that are really requirements

Can the assistant diagnose?

No. It routes to a specialty and flags emergencies; diagnosis stays with the doctor.
So: a red-flag gate runs before any match, and wording guardrails sit on every patient answer.

Can a patient share records with family?

Yes, through a caregiver role whose consent expires.
So: consent is a table with scope, grantee and expiry, never a checkbox.

Where does the consultation audio go?

Processed and deleted within 24 hours; the transcript and the note are kept.
So: an object store with a lifecycle rule, and deletion writes an audit event.

May identified data reach a public LLM?

Only de-identified text may cross the hospital boundary.
So: a scrubber sits on the egress path and identified text stays in-region.

Where does stock data come from?

A nightly CSV today, an API in about six months.
So: one stock interface with two adapters, and the graph never knows which is live.

Who approves AI-drafted notes?

The consulting doctor, every time; nothing auto-signs.
So: the graph interrupts before persistence, and an unsigned note is a draft by schema.

What volume should we plan for?

9,000 appointments a day; 1,400 requests an hour at the morning peak.

So: load-test at peak, ingest asynchronously, keep the API stateless behind a balancer.

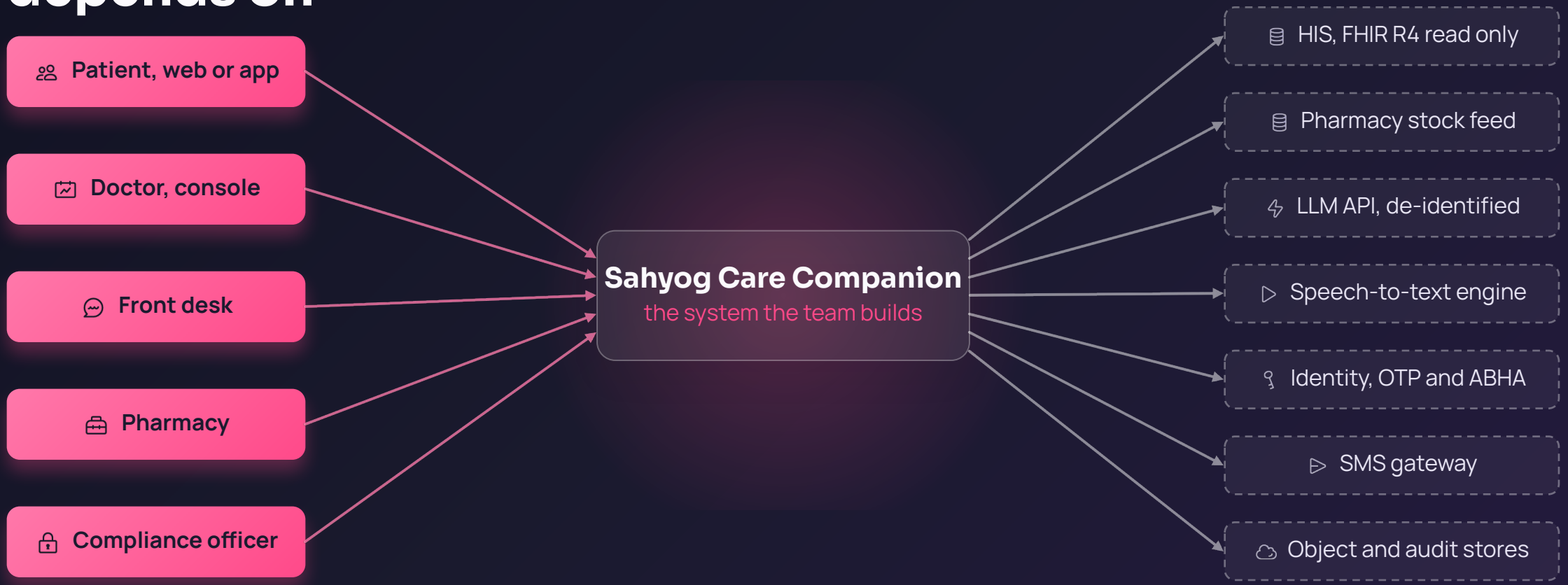
How to read an addendum

Every answer is a requirement wearing a conversational tone. The team's PRD-lite cites the addendum line behind each design rule, so the client reads its own words in the design.

DESIGN

Context view: who touches the Care Companion, and what it depends on

People sit on the left, dependencies on the right, and every arrow follows the direction of the request.



CONTEXT

Stakeholder map: who can stop the engagement, and the first proof each is owed

High power, lower interest: keep satisfied

- CIO and IT head
- Finance, not in the RFP

High power, high interest: manage closely

- Chief medical officer
- Compliance officer
- Doctors

Low power, low interest: nobody the RFP names sits here

Low power, high interest: keep informed

- Nursing lead, not in RFP
- Front desk
- Patients
- Pharmacy

CMO fears AI advice harming a patient; owed the red-flag gate, shown on the hard cases.

CIO fears lock-in and a leak; owed the container view with the egress boundary marked.

Compliance fears a DPDP breach in month one; owed the audit log and the consent table.

Doctors fear being blamed for an AI note; owed proof that nothing persists unsigned.

Front desk fears replacement; owed a flow that escalates to them, with a queue view.

Pharmacy fears substitutions made by software; owed stock flags with the choice left human.

Patients fear strangers reading their reports; owed the scope guard and citations on every answer.

DELIVERABLE D1

The automation ladder: every client task placed on a rung before any build

FOUR QUESTIONS DECIDE THE RUNG

- Is the rule fully known and stable?
- Is the input bounded and well formed?
- What does a wrong answer cost, and who notices it?
- Who must sign off before the effect is real?

Two “yes” answers at the top put a task on the code rung; a cost measured in patient harm or in law puts it on the human rung whatever the model can do. The ladder is a programme device, coined for this engagement.

4 Human decides
the model may list options
- Sign the clinical note
- Pick a substitute medicine

3 Agentic loop
the path depends on tool results
- Choose among six candidate doctors

2 LLM-augmented
one fuzzy step inside checked code
- Symptoms to a specialty
- Answer a question on a report
- Dictation to a SOAP note
- Does this question need a doctor
- Red flags: rules first, LLM second

1 Deterministic code
known rule, bounded input
- Is the slot still free
- Send the confirmation
- PDF to text
- Medicine in stock

D1 is accepted when the twelve client tasks above and at least six the team found in discovery each sit on one rung with the four answers recorded.

DELIVERABLE D2

Ten discovery questions, each tied to the decision its answer unlocks

1. Which specialties are misrouted most today, and how do you know?
   Unlocks the first eval cases and the crosswalk.
2. What must happen when a patient reports chest pain at night?
   Unlocks the red-flag list and the escalation path.
3. Who may read a record besides the patient, and for how long?
   Unlocks the consent model and the access matrix.
4. Where may identified data be stored and processed today?
   Unlocks on-premise, private or hybrid.
5. Which system is the source of truth for the appointment book?
   Unlocks whether you write bookings or propose them.
6. What does a doctor do when the draft note is wrong?
   Unlocks the shape of the interrupt and the edit loop.
7. How is pharmacy stock refreshed, and how stale may it be?
   Unlocks cache lifetime and the CSV versus API adapter.
8. What is the budget per consultation for AI services?
   Unlocks model routing and the small-model share.
9. Who owns an incident when the assistant misleads a patient?
   Unlocks the audit trail, review queue and escalation owner.
10. What would make the CMO switch this off in week one?
    Unlocks the first demo and the first guardrail.

D2 is accepted when the team's own ten questions are posted in the client thread, at most three of them taken from this page; faculty answer in role as the client within one working day, and each answer lands in the assumptions register.

DELIVERABLE D3

The assumptions register, and what closes discovery

Assumption	How the team validates it	Fallback if it fails
The HIS FHIR API answers a demographics read within 300 ms.	Time fifty calls against the sandbox, or a mock with injected latency.	Cache demographics per session and degrade to manual lookup.
The nightly stock CSV is complete before the clinics open.	Check file timestamps across a week of samples.	Show stock age on every flag; mark unknown when stale.
Patients type symptoms in English or mixed Hindi-English.	Generate 200 synthetic intakes in both styles and run them through matching.	Add a language-detection node before the specialty match.
A doctor reviews the draft during the consultation itself.	Role-play five consultations with faculty acting as the doctor.	Queue drafts for an end-of-session review instead.
One slot maps to exactly one appointment.	Ask the front desk whether overbooking is practised.	Model capacity per slot instead of a free-or-taken flag.
De-identified text is enough for a good SOAP draft.	Compare drafts on thirty transcripts with and without identifiers.	Keep identified drafting on-premise with a local model.

Discovery closes when

- D1 holds the twelve client tasks plus six the team found, each on a rung.
- D2's ten questions are posted and answered in the client thread.
- D3 has at least eight rows, each with an owner, a validation and a fallback.
- The data inventory records exact row counts and licences for every source.
- The repository exists with the structure on page 48.

Five people the system must serve, and the one thing each must get

Priya, 34

Patient, who also manages her mother's care

Needs the right specialist tonight and yesterday's report in plain words.

Fears her mother's reports being read by someone who should not see them.

Dr. Rao

Cardiologist, 22 patients a day

Needs a note he can sign in ninety seconds without re-reading the transcript.

Fears being on record for a line the model invented.

Meena

Front desk lead, 300 calls a day

Needs fewer calls about which doctor to see, and a queue she still controls.

Fears a bot that books the wrong slot and sends the patient to her counter.

Imran

Pharmacist, two counters

Needs prescriptions that match stock before the patient walks over.

Fears substitutions decided by software instead of by him.

Kavita

Compliance officer

Needs to answer "who saw this record" with one query.

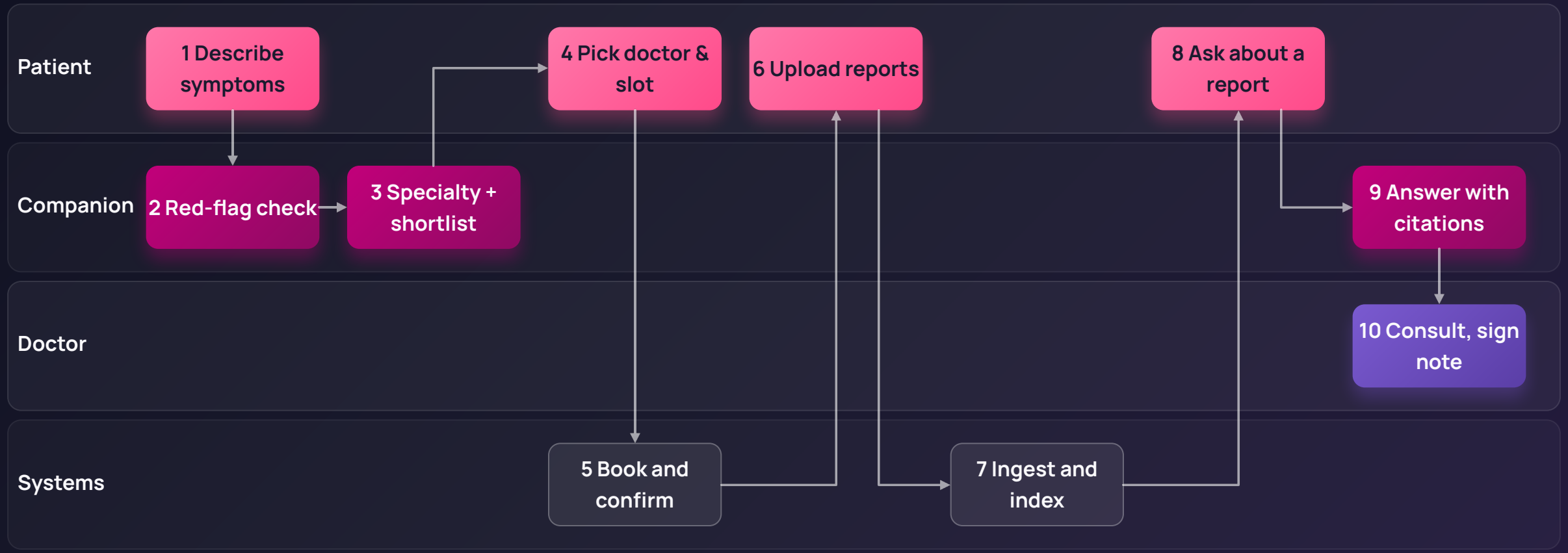
Fears identified health data in a public API log.

Every epic on page 15 names the persona it serves; a feature that serves nobody on this page is scope creep, and the client is not paying for it.

DESIGN

Patient journey: from symptoms to a signed note in ten steps

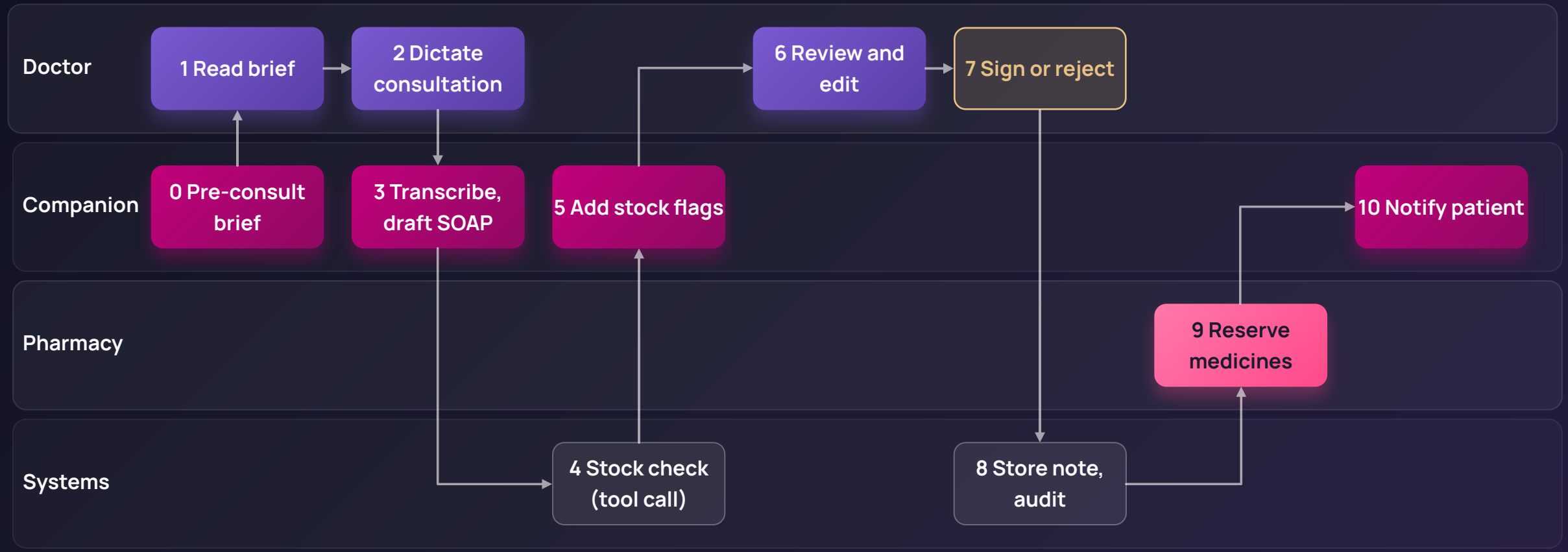
Hand-offs stay in one column; a step moves right only when the same lane continues. Pink is the patient, magenta the Companion, violet the doctor, glass the systems.



DESIGN

Doctor journey: dictation to a signed, stock-aware note

A rejection at step 7 returns the draft to step 3 with the doctor's comments; nothing is stored before the signature. Violet is the doctor, magenta the Companion, pink the pharmacy, glass the systems.



DESIGN

Two real messages, and what the Companion must do with each

CASE A · RED FLAG

"Hi, my mother (62) has had a dull ache in her left arm and some breathlessness since this morning. She has diabetes. Which doctor should we see?"

1 rules fire — Arm pain plus breathlessness, age over 60, diabetes: the deterministic rule set marks a red flag before any model is called.

2 hand-off — The graph leaves the booking path, opens an emergency-desk ticket with the patient's contact, and ends the run.

3 reply — "These symptoms need urgent attention now. Please call 112 or go to the nearest emergency department. I have alerted Sahyog's emergency desk; a nurse will call you within two minutes."

No diagnosis is named, the audit row records which rules fired, and the eval set holds this case with red_flag = true as a hard requirement.

CASE B · ROUTINE

"Dull lower back pain for three weeks, worse after long sitting. No fever, no weakness in the legs."

1 rules clear — No red-flag rule matches, so the specialty-match node runs the model against the validated crosswalk.

2 output

```
{ "red_flag": false, "specialty":
"Orthopaedics", "alternative":
"Physiotherapy", "confidence": 0.82,
"candidates": 3, "next_slot_in": "18
h" }
```

3 interrupt — Three doctors with their next free slot go to the patient; the run pauses in the checkpointer until the patient picks one.

A confidence under 0.6 triggers one clarifying question, never a guess; two low-confidence turns hand the patient to the front desk.

Six epics, and the acceptance criteria the client will test against

Epic, and the persona served	Acceptance criteria, testable as written
E1 Find my doctor (Priya)	Free-text symptoms return one specialty and three doctors with their next free slot within four seconds; any red-flag phrase ends the flow with emergency guidance and a logged human hand-off.
E2 Book and manage (Priya, Meena)	Booking is idempotent under retries, confirms by SMS, and the front desk can see and override any booking the assistant made.
E3 My records (Priya)	A text PDF is stored against the patient within sixty seconds with a visible status; a caregiver sees it only while consent is valid.
E4 Ask my record (Priya)	Every answer cites the document and page it came from, refuses questions outside the patient's own records, and hands off to a doctor when asked for a diagnosis or a dose.
E5 Scribe (Dr. Rao, Imran)	A transcript becomes a SOAP draft with medicines flagged against stock; the draft is persisted only with the doctor's signature and its edit history.
E6 Platform (Kavita)	Every record read writes an audit event with actor and purpose; consent carries scope and expiry; identified data never reaches an external API.

Each criterion is written so a test can pass or fail it; the team's PRD-lite breaks every epic into user stories with the same discipline and cites the RFP line behind each one.

ASSURANCE

Non-functional targets, and the numbers the demo has to put on screen

Requirement	Target
Triage turn latency	p95 under 4 seconds
Record answer latency	p95 under 8 seconds
Note draft latency	p95 under 20 seconds after the transcript
Availability	99.5% during clinic hours
Audit	Every record read logged with actor, purpose, time
Residency	Identified data stored and processed in India
Retention	Audio gone within 24 hours; audit kept one year
Recovery	Restore within 4 hours; checkpoints survive restarts

Metric on the demo screen	Target
Bookings completed without a call	At least 70% of synthetic sessions
Time to book	Median under 3 minutes
Red-flag recall on the eval set	100%, with no exceptions
Specialty accuracy	At least 85% on the golden set
Answer faithfulness	At least 0.9 by a calibrated judge
Scope leaks	Zero across the red-team set
Doctor review time per note	Under 90 seconds in role-play
Cost per consultation	Computed, with the formula shown

These are the client's asks. A target the team cannot meet is reported with the number reached and the reason, never dropped from the slide; p95 excludes the slowest five percent and is the figure operations teams quote.

CONTEXT

The data estate: five public sources, and one layer the team generates

Medical Appointment No Shows

Kaggle, by Joni Hoppen (Aquarela); OpenML id 43439, CC BY-NC-SA 4.0

110,527 rows

Scheduling history and the no-show signal behind slot logic.

Medical Transcriptions

Kaggle: tboyle10/medicaltranscriptions

about 5,000 notes

Seed for dictation transcripts and the synthetic report PDFs.

Disease Symptom Prediction

Kaggle: itachi9604/disease-symptom-description-dataset

about 4,900 rows

Symptom-to-condition signal behind the specialty match.

A-Z Medicine Dataset of India

Kaggle: shudhanshusingh/az-medicine-dataset-of-india

250,000+ medicines

Pharmacy formulary for stock checks and substitutes.

Healthcare Dataset, synthetic

Kaggle: prasad22/healthcare-dataset

about 55,000 admissions

Patient master and encounter history.

Generated layer, the team's own

Produced by versioned prompts in the repository

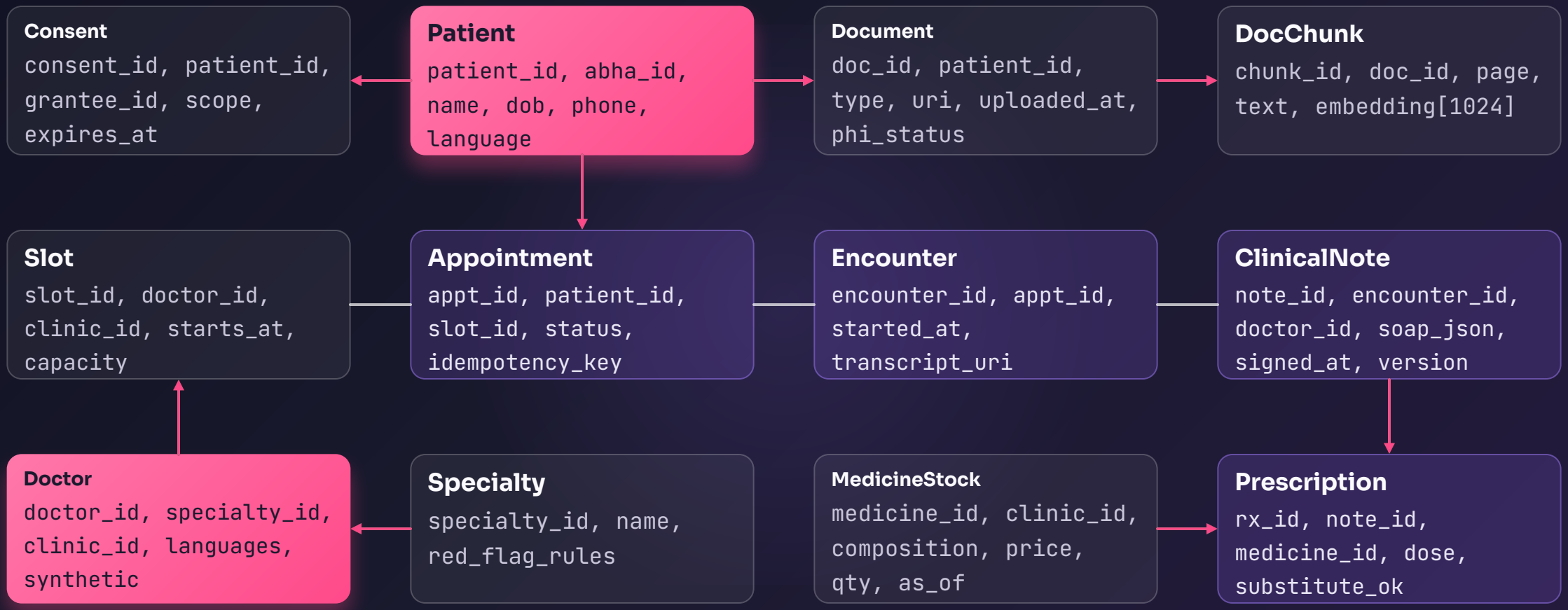
500 doctors, 480k slots

Plus 2,000 report PDFs and 300 transcripts: everything the public sets lack.

Existence of each dataset was verified on 21 October 2026; sizes marked "about" are from memory, and the team's data inventory records the exact row count and licence before any row is used.

DESIGN

Entity model for Postgres: arrowheads mark the many side, plain lines are one-to-one



DESIGN

Cleaning and transformation, source by source, each with the proof it worked

Source	Known mess	Transformation	The check that proves it
No Shows	Negative ages; schedule dates after the visit date; a target column that reads backwards.	Parse dates, drop impossible rows with a tally, map areas to the fictional clinics, rename the target to attended.	Counts before and after, with the drop reasons tallied.
Transcriptions	Template headers inside the body; forty uneven specialty labels; duplicates.	Normalise to the 38 specialties, strip headers, scrub names with Presidio.	Fifty notes read by eye for leftover identifiers.
Disease Symptom	Wide symptom columns with stray underscores; conditions with no specialty.	Melt to long form; generate, then hand-check, the condition-to-specialty crosswalk.	Every crosswalk row carries a reviewer mark.
A-Z Medicine	Price strings with symbols; brand duplicates per composition.	Parse prices, group by composition, extract pack size by regex with an LLM fallback.	Regex and LLM agree on 95% of a 200-row sample.
Healthcare Dataset	Random names; discharge before admission; coarse test results.	Use as encounter history only; fix the date order; map hospitals to the network.	Date-order violations fall to zero.
Generated layer	Repetitive names; implausible calendars; a skewed specialty mix.	Seed names from a distribution; bound calendars by clinic hours; audit the mix.	A share table against the 1,400-doctor split.

Each check becomes a test under data/transforms/tests. Two tables the team adds itself: Clinic, which owns slots and stock, and AuditEvent, one row per read or write of any table in the model.

DESIGN

The generated layer: what an LLM writes for the team, and how it is kept honest

Doctor directory, 500 rows
Name, specialty, clinic, languages and years in practice, written by the LLM against the 38-specialty split.

Slot calendar, about 480,000 rows
Sixty days of sixteen slots per doctor, built by plain code from clinic hours; a calendar is arithmetic, so no LLM touches it.

Documents and dictations
2,000 lab and radiology PDFs rendered from LLM-written text seeded by the transcriptions, plus 300 dictation transcripts; the Pro tier adds TTS audio and runs it back through speech-to-text.

1. Seed from real columns, so specialties, drug names and lab test names come from the public sets, never from the model's imagination.
2. Prompt with a JSON schema and three few-shot rows; the schema is the same pydantic model the ingest code uses.
3. Generate in batches of fifty with a fixed seed value in the prompt, so a failed batch reruns without shifting the rest.
4. Validate every row with the pydantic rules and reject, never repair, a row that fails; rejected rows are counted in the inventory.
5. Audit for duplicates and bias: name repetition, gender and language mix per specialty, and clinics with no doctors.
6. Version the prompt, the seed and the output together, and tag every generated row with synthetic = true so it never mixes unlabelled with public rows.

DELIVERABLE D4

Data contracts, and the four classes that decide what may cross the boundary

One contract per table carries

- An owner, named as a role and never as a person.
- A schema version and the rule for changing it.
- Freshness: how stale a row may be before a consumer must know.
- A sensitivity class from the table on the right.
- Quality checks for nulls, ranges and uniqueness, each with a threshold.
- Its consumers, and what breaks for each when the schema changes.

D4 is accepted with fourteen contracts filed under docs/data-contracts and the class column signed off by the compliance officer in role.

Class	Examples	Leaves the boundary?	Treatment
Direct identifiers	Name, phone, ABHA number, photo	Never	Tokenise before any external call; Presidio 2.2.x finds the common ones, and the team's own recognisers catch Indian phone and ABHA formats.
Quasi-identifiers	Date of birth, pincode, a rare condition	Only generalised	Send an age band instead of a birth date and a district instead of a pincode.
Clinical content	Diagnosis text, medicines, lab values	Only de-identified	Scrub, then send; keep the token map on-premise so the answer can be re-identified for the patient.
Operational	Slot times, specialty names, stock counts	Yes	Plain handling, still written to the audit log when tied to a patient.

DELIVERABLE D5

Is there enough data? The sizing the team redoes with its own counts

110k appointments

250k+ medicines

55k encounters

480k generated slots

60k chunks from 2,000 PDFs

5k transcripts

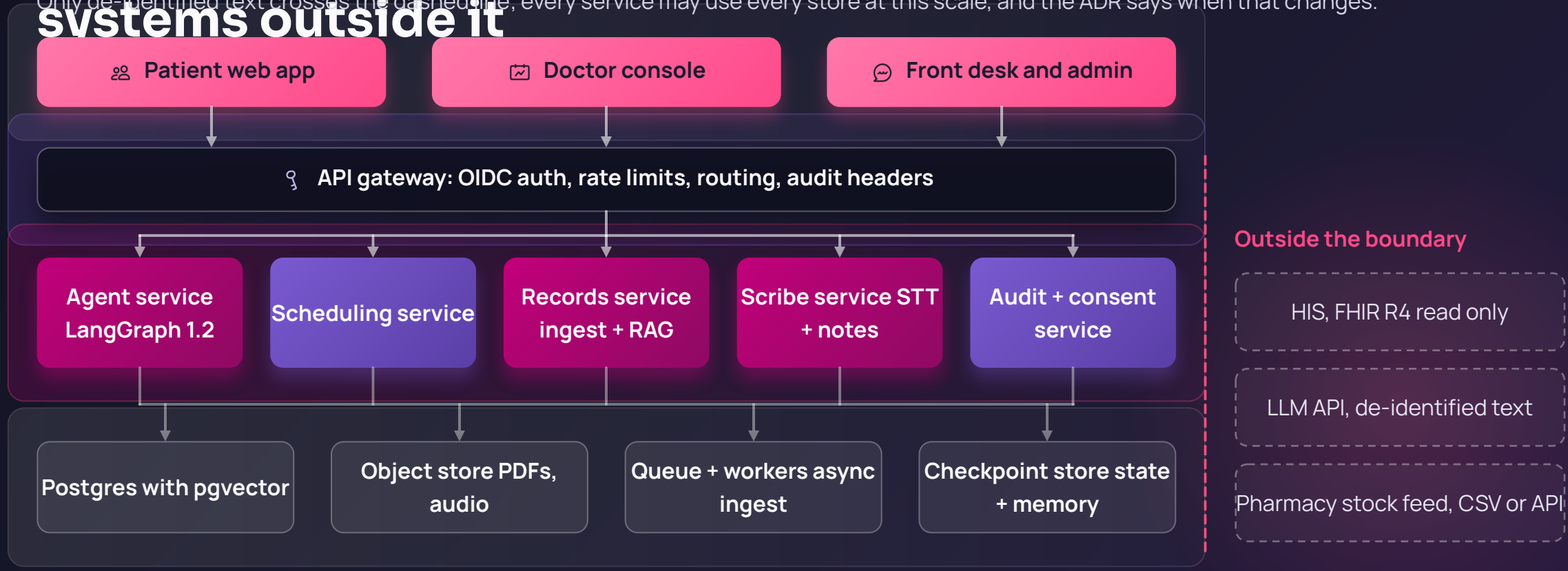
Store	Rows	Size rule	Estimate
Vector index	60,000 chunks, 2,000 PDFs at about 30 each	rows × 1,024 dimensions × 4 bytes	About 246 MB in float32; half of it in float16
Chunk text	60,000 chunks	About 1,200 characters a chunk	About 72 MB
Slot calendar	480,000 rows	About 80 bytes a row	About 38 MB
Checkpoints	9,000 runs a day × about 8 steps	A few kilobytes a step	Tens of MB a day, so a retention rule is required
Audio	9,000 consultations × about 8 minutes	About 1 MB a minute compressed	About 70 GB a day, deleted within 24 hours

Enough means: every eval case has at least five candidate chunks, every specialty has at least ten doctors, and every clinic has slots on every working day. D5 is the inventory with these three checks passing.

DESIGN

Container view: four tiers inside the hospital boundary, three systems outside it

Only de-identified text crosses the dashed line; every service may use every store at this scale, and the ADR says when that changes.



DESIGN

Inside the agent service: entry, graph runtime, tools, and what cuts across all three

1 Entry

- One FastAPI router per workflow, so W1, W2 and W3 can scale apart later.
- An auth context on every request: who, which role, which consent scope.
- Request validation with pydantic, the same models the data layer uses.
- An idempotency key on every write, generated by the client.

2 Graph runtime

- Three compiled LangGraph graphs, one per workflow, built once at start-up.
- A Postgres checkpointer (langgraph-checkpoint-postgres 3.2.x) keyed by thread id.
- A long-term store keyed by patient for preferences that outlive a thread.
- The interrupt-and-resume handler that pauses a run for a human.
- Node-level timeouts and retries, so one slow tool cannot stall a run.

3 Tools

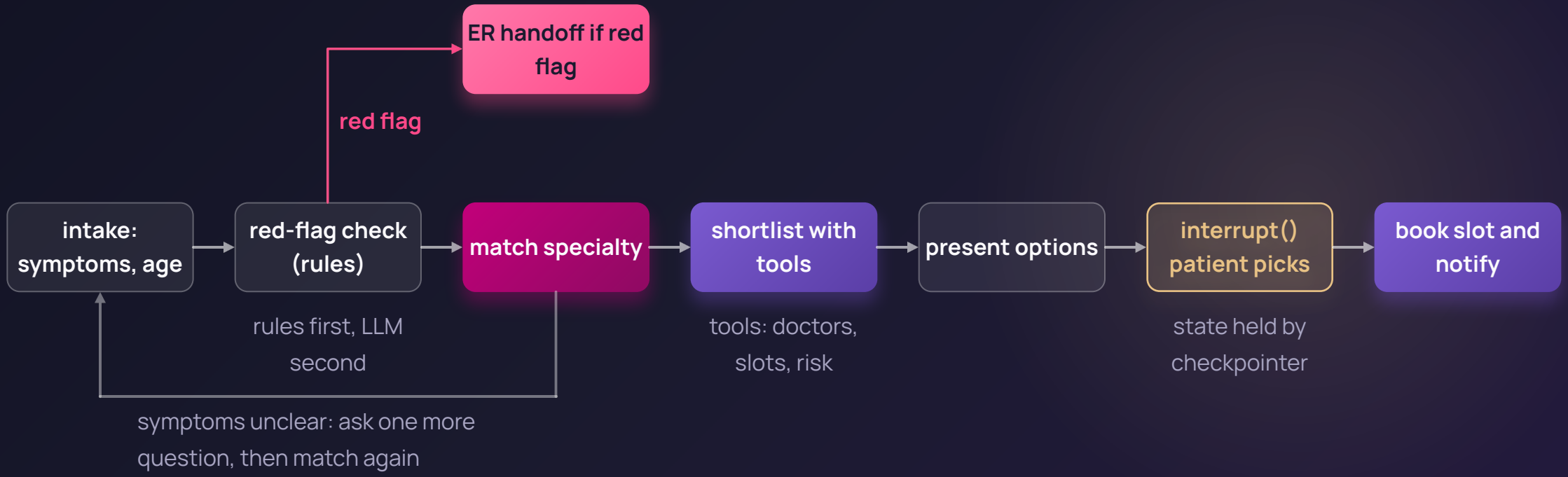
- Scheduling client: reserve and release a slot.
- Records retriever: hybrid search with a hard patient filter.
- Stock lookup behind one interface, with CSV and API adapters.
- FHIR reader for demographics and encounters.
- Speech-to-text client and the SMS notifier.

On every call: trace id and spans | PII scrubber on egress | guardrails in and out | rate limiter | cost meter per run

DESIGN

W1 Find my doctor: the first graph the team ships

Glass nodes are plain code, magenta nodes call a model, violet nodes call a tool, gold waits for a human. State: messages, symptoms, age, red_flags, specialty, candidates, chosen_slot, booking_id, audit_ids.

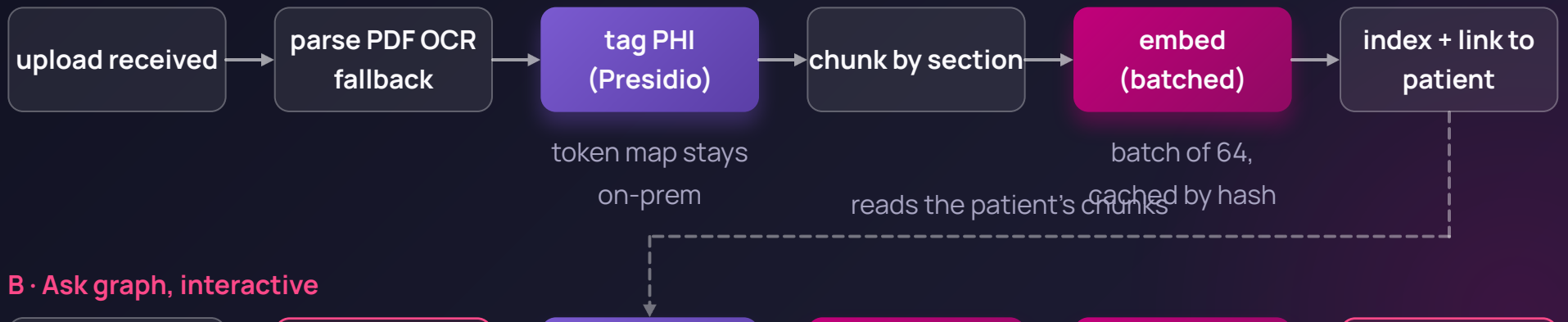


06 · SOLUTION ARCHITECTURE

DESIGN

W2 My records: an ingest graph on the worker, an ask graph on the request

A · Ingest graph, on the queue worker



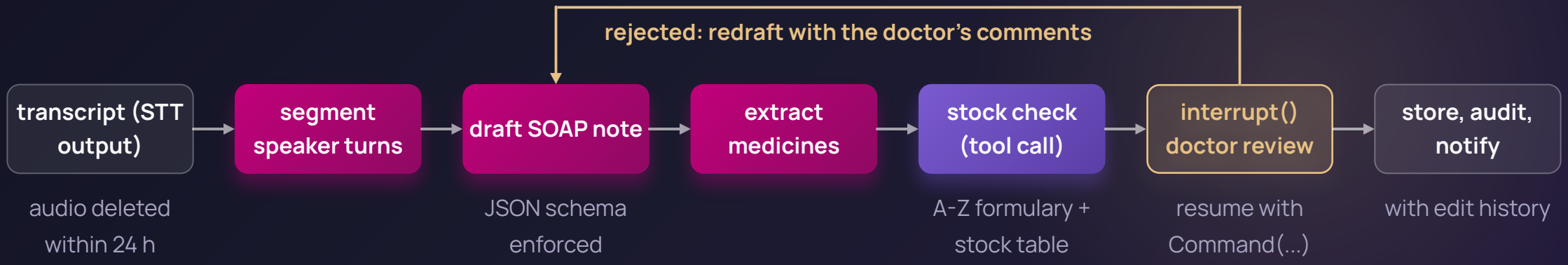
B · Ask graph, interactive



DESIGN

W3 Consultation scribe: a graph that pauses for a signature

Glass nodes are code, magenta nodes call a model, violet nodes call a tool, gold waits for the doctor. State: transcript, turns, soap_draft, medicines, stock_flags, doctor_decision, note_id, audit_ids.



Nothing in this graph writes to the record before node 6 returns the doctor's decision; a timeout on the interrupt parks the draft in the review queue instead of failing the run.

DESIGN

A worked note: what the scribe drafts, and what the doctor changes before signing

TRANSCRIPT EXCERPT, SYNTHETIC

Doctor: So the tightness in the chest comes on when you climb stairs and settles after a few minutes of rest?

Patient: Yes, two or three times this week. Nothing at rest.

Doctor: Blood pressure is 138 over 86 today. I'll start you on a low-dose statin, and I want an ECG and a lipid panel before we meet again in two weeks.

The doctor named a class, not a molecule. What the model does with that gap is the whole test of this workflow.

DRAFT FROM THE GRAPH, 11 SECONDS

```
S Exertional chest tightness, 2 to 3 episodes this week, settles with
rest; none at rest.
O BP 138/86. ECG and lipid panel ordered.
A As stated by the doctor: exertional symptoms, cardiac work-up
initiated.
P Statin, low dose [molecule and dose not stated: doctor to specify];
ECG; lipid panel; review in two weeks.
```

Doctor edits one field: selects atorvastatin 10 mg nightly. The stock tool runs on resume: in stock at the Baner pharmacy, 240 units.

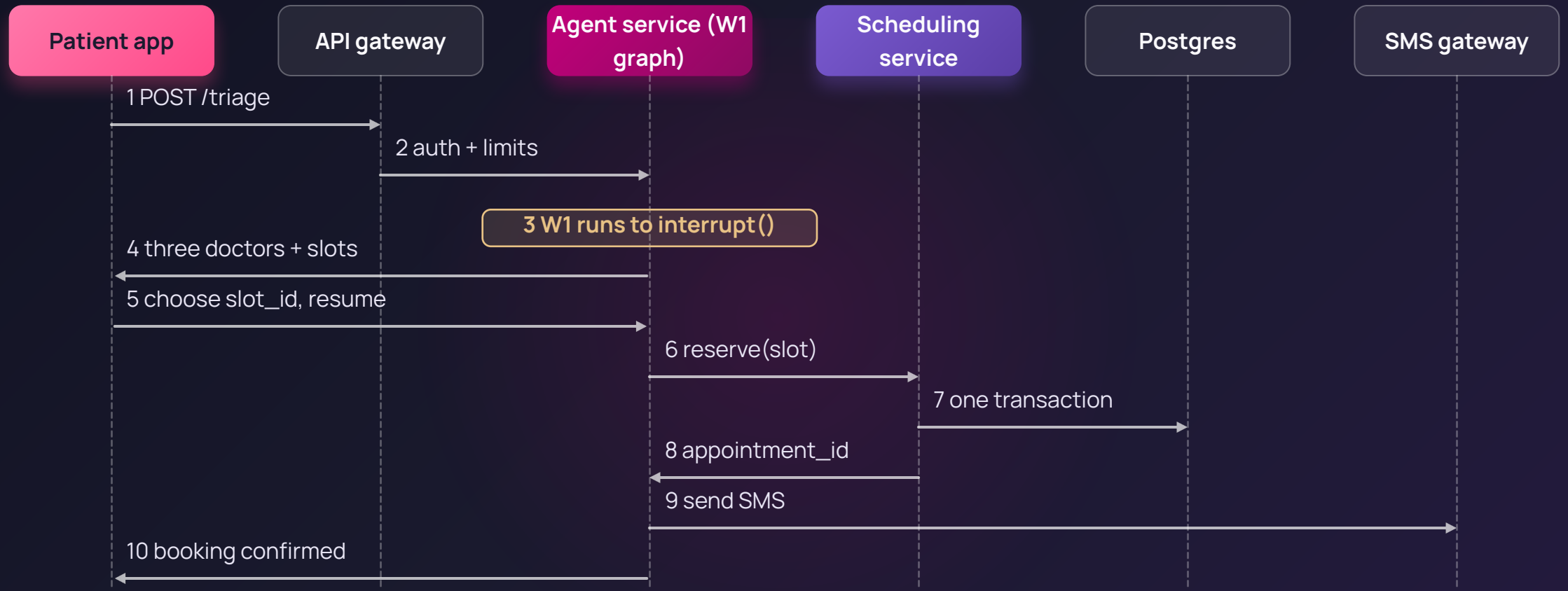
Signs after 70 seconds. Note version 1 persisted with the edit history; audit event written; patient notified of the two tests.

Every line in A and P quotes the doctor; the model structures, it never prescribes. A draft that fills the molecule itself fails the eval set, whatever the molecule.

DESIGN

Sequence: booking a slot through W1 across six components

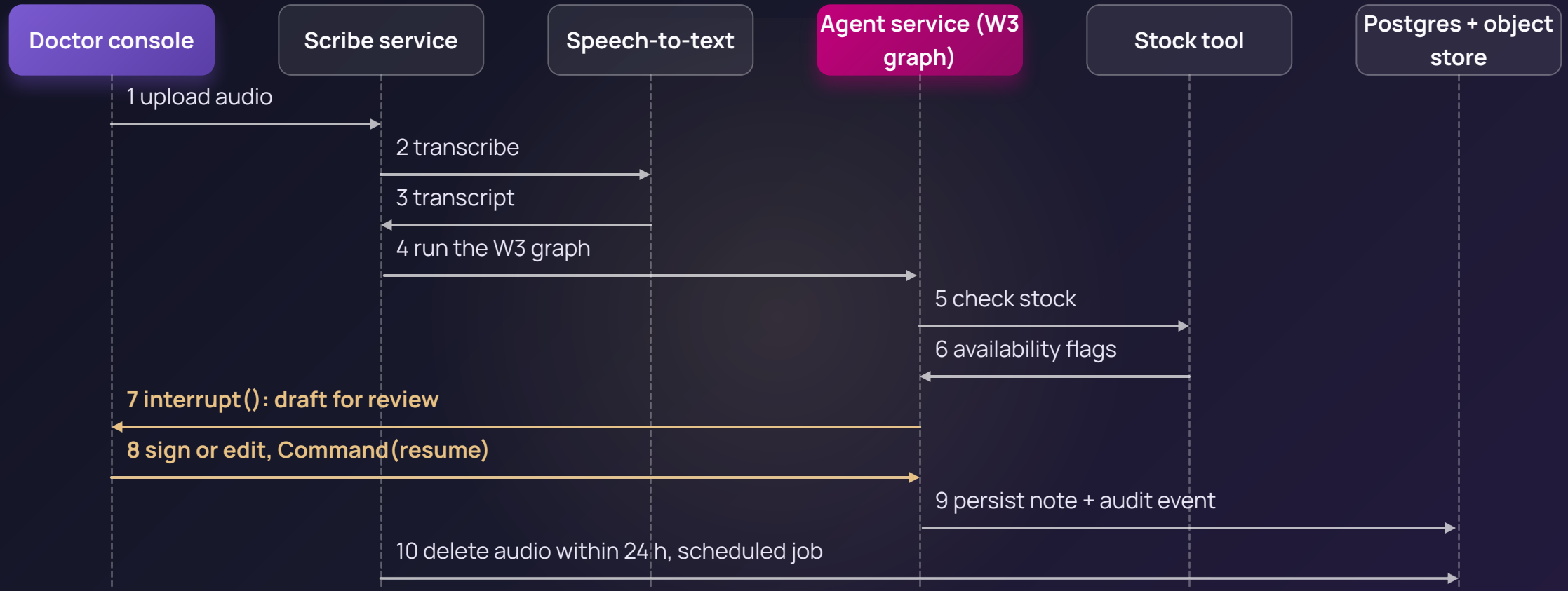
Ten messages. The run pauses at message 3 and resumes at message 5 from the saved checkpoint, so the API itself stays stateless.



DESIGN

Sequence: a consultation becomes a signed note

The doctor's signature at message 8 resumes the paused run; the audio deletion at message 10 is a scheduled job, not a graph step.



DESIGN

Memory has four homes, and guardrails have three layers

Memory	Lives in	Example	Expires
Thread state	The checkpointer, in Postgres	The triage conversation and its chosen slot	Kept 30 days for replay, then purged
Long-term store	A key-value store keyed by patient	Preferred language, usual clinic	When the patient changes it
Retrieval memory	pgvector chunks	Report contents, with citations	With the document, under consent
Operational cache	Redis	Slot availability	After 30 seconds

Input layer
Scrub identifiers from anything leaving, scan uploaded text for injected instructions, and take the scope from the auth context, never from the prompt.

Model layer
Constrain outputs to a schema, forbid diagnosis and dosing in the system instruction, and allow each graph only the tools it needs.

Output layer
Check every citation against the retrieved chunks, escalate red flags, and require a human signature for any clinical act.

Guardrails are code and tests, not sentences in a prompt: each one has a red-team case in the eval set that fails when the guardrail is removed. A prompt can be argued with; a SQL filter cannot.

DELIVERABLE D6

Where it runs: four deployment options scored against the client's constraints

Criterion	On-premise	Private cloud, India region	Public cloud + public LLM	Hybrid
Identified data residency	Strong: never leaves the site	Strong: India region by contract	Weak: depends on vendor terms	Strong: identified data stays in
Access to current models	Weak: only what you can host	Fair: in-region hosted models	Strong: any model, same day	Strong: de-identified calls only
Operations burden	Weak: you run everything	Fair: managed data, your containers	Strong: managed end to end	Fair: two estates to run
Cost at 9,000 a day	Fair: high fixed, low variable	Fair: steady and predictable	Fair: low fixed, variable grows	Fair: variable on LLM calls only
Time to first value	Weak: procurement first	Fair: weeks	Strong: days	Fair: days to demo, weeks to harden
Audit and control	Strong: full	Strong: contractual	Fair: shared responsibility	Strong on identified data

D6 is an architecture decision record: the team rescores this matrix with the Gate 2 answers, defends one option, and names the trigger that would make it switch. The scores above are the delivery office's opening view, not the answer; pink reads strong, gold fair, grey weak.

DESIGN

Who may do what: the access matrix, consent and the audit rule

Resource	Patient	Caregiver	Doctor	Front desk	Pharmacist	Compliance
Own record	Read	Read while consent is valid	Read if treating	None	None	Audit only
Any patient record	None	None	Read if treating	Demographics only	None	Audit only
Appointment book	Own only	Own ward's	Own slots	Read and write	None	Read
Clinical note draft	None	None	Write and sign	None	None	None
Signed note	Read own	Read with consent	Read	None	Prescription lines	Read
Audit log	Own events	None	None	None	None	Read all
Stock data	None	None	Read	None	Read and write	Read

Consent is a row, not a checkbox: grantee, scope, purpose and expiry, checked on every read.

Every record read writes an audit event with actor, role, purpose, record and time; the compliance officer's one query is a SELECT on that table.

Identity comes from an OIDC provider with OTP; linking an ABHA number is an Architect-tier extension, not a Core requirement.

DELIVERABLE D7

Threat model: five ways this system hurts someone, and the proof it cannot

Flow	Threat, with its STRIDE class	Control	How the team proves it
Ask my record	Instructions hidden in an uploaded PDF make the assistant read another patient's chunks (information disclosure).	Patient filter applied in SQL at retrieval; output scanned for foreign document ids.	A red-team PDF in the eval set; the test fails when the filter is removed.
Find my doctor	Scripted bookings hoard slots (denial of service).	Rate limit per patient and per address, OTP at booking, idempotency keys.	A load script tries 1,000 bookings and gets twenty.
Scribe	A note is edited after the signature (tampering).	Signed notes are immutable; an edit creates a new version with a hash in the audit event.	An update attempt returns a rejection and leaves an audit row.
Egress	Identified text ends up in a vendor's log (information disclosure).	Scrubber on the egress adapter, token map on-premise, zero-retention terms with the vendor.	Replay 200 outbound payloads through a detector and get zero hits.
Data at rest	A database dump walks out (information disclosure).	Encryption at rest, least-privilege roles, no shared admin account.	An access review lists every role and its grants.

STRIDE names six threat classes: spoofing, tampering, repudiation, information disclosure, denial of service and elevation of privilege. D7 adds one row per class not covered here, and the red-team PDF is the single most valuable test in the engagement.

ASSURANCE

Cost per consultation: a worksheet the client can audit, never a quoted price

Cost line	Driver the team measures	Formula	Source of the number
LLM tokens, triage	About six turns, 1,500 tokens in and 300 out per turn	turns × (in × p_in + out × p_out)	Traces; the provider's price list on the day
LLM tokens, record answers	About 4,000 tokens in and 400 out per answer	answers × (in × p_in + out × p_out)	Traces
LLM tokens, scribe	About 3,000 tokens in and 1,000 out per note	notes × (in × p_in + out × p_out)	Traces
Speech-to-text	About eight minutes of audio per consultation	minutes × p_minute	Provider price list
Embeddings	New chunks per day × tokens per chunk	tokens × p_token	Ingest logs
Storage and compute	Gigabytes stored; worker hours	GB × p_gb + hours × p_hour	Cloud bill, or on-premise amortisation
Human review time	Doctor seconds per note	(saved − spent) × loaded rate	Role-play timings

Cost per consultation is the total divided by consultations. The team compares it with the eleven minutes the RFP wants back, records the price-list date beside every number, and never hard-codes a price in the repository; model routing on the next page is how the triage line moves from a large model to a small one.

DESIGN

Eight production concerns the demo must at least point at

Authentication
OIDC with short-lived tokens; the auth context travels with the request into graph state, so a tool never trusts a patient_id the model typed.

Rate limiting
Token buckets at the gateway per patient and per address, with a separate, stricter bucket for booking writes.

Async and long-running
Ingest runs on a queue with a job id and a status endpoint; the client polls or subscribes and never holds an HTTP connection open for a PDF.

Parallelism
Fan out summarisation across documents with LangGraph's Send API and a bounded worker pool, then join before the answer node.

Model routing
A small model classifies and extracts, a large model drafts, and a fallback provider is configured and tested rather than merely named.

Retries and timeouts
Every tool call has a timeout, a capped exponential backoff and a circuit breaker; when the LLM API is down, the run degrades to the human queue.

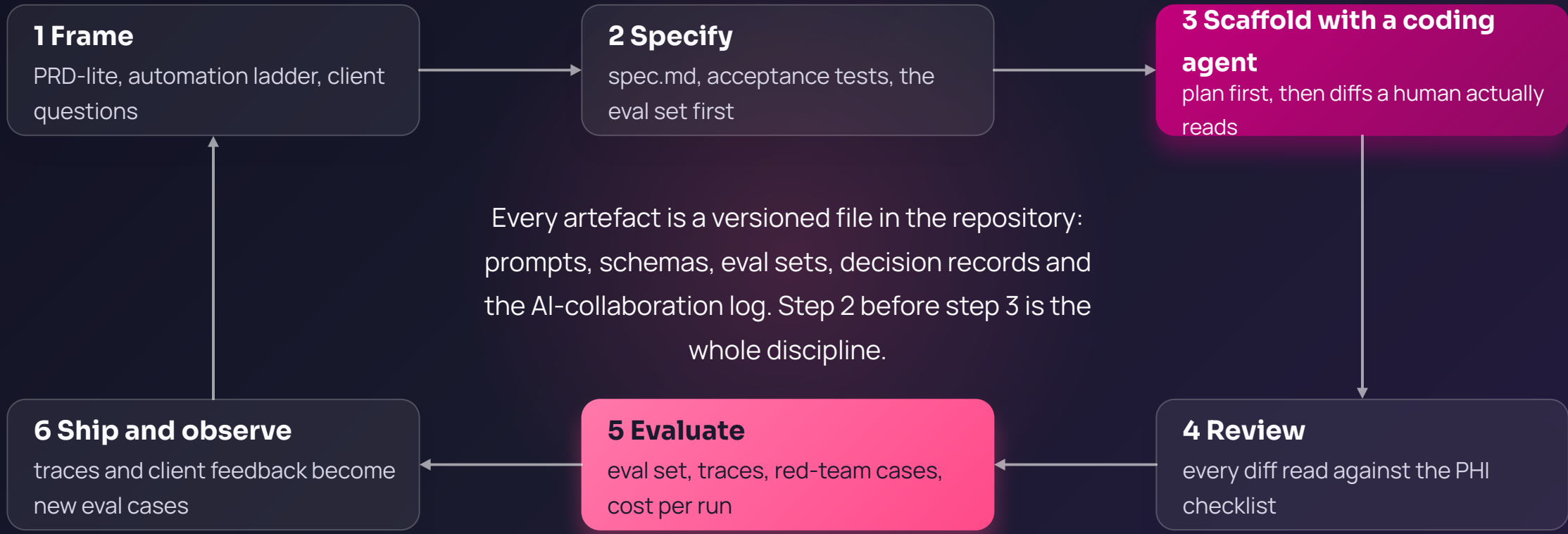
Caching
Slot availability for thirty seconds, embeddings by content hash, and prompt prefixes wherever the provider supports them.

Load balancing
Stateless API containers behind a balancer; all run state lives in the checkpointer, so any replica can resume any thread.

Core tier shows the first three working; Pro adds routing, retries and caching; Architect proves parallelism and balancing under the 1,400-an-hour load test. Not all eight are in the taught syllabus, and an FDE meets all eight on the first client call.

DESIGN

The agentic development loop: once per epic, with evals written before code



Every artefact is a versioned file in the repository: prompts, schemas, eval sets, decision records and the AI-collaboration log. Step 2 before step 3 is the whole discipline.

CONTEXT

Ten house rules for building with models, binding on every pod

1. Write the spec and the eval set before the first prompt; a prompt without a test is a guess.
2. Prompts, schemas and eval sets live in the repository and change only through pull requests.
3. Every pull request carries an AI-collaboration note: what the agent proposed, what was rejected, and why.
4. A coding agent never sees real identified data; it works on the generated layer only.
5. Deterministic first: when the rule is known, write code and keep the model for the fuzzy step.
6. Every model call has a schema, a timeout, a fallback and a trace id, or it does not ship.
7. Read the trace before changing the prompt; most “prompt problems” are retrieval or state problems.
8. Any clinical or financial effect waits for a named human; the graph pauses, it never guesses.
9. Keep a decisions log of ADRs; the steering review walks through it, not through the code.
10. When the model surprises the team, the case goes into the eval set the same day.

Rule 3 is the pass-or-fail gate in the scoring: a repository with no AI-collaboration notes is not scored, whatever the code does. These rules are the delivery office’s own, coined for this programme.

CONTEXT

Thirty minutes to research a library, and what this pack verified

1 **Official docs** for the current major version, and the version read is written down.

2 **Changelog and migration guide** between that version and the one in the lock file.

3 **Open issues** sorted by reactions, for the sharp edges the docs leave out.

4 **A minimal repro** in a scratch file, under five minutes, proving the behaviour relied on.

5 **Decide and record** in an ADR with the version pinned and the date it was checked.

Package, checked on PyPI 21 Oct 2026	Version	Why it matters here
langgraph	1.2.16	Graphs, interrupts, checkpoints
langgraph-checkpoint-postgres	3.2.0	Durable thread state
mcp	2.3.0	The MCP server extension
langfuse	5.0.3	Tracing; LangSmith is the other option
ragas	0.4.3	Faithfulness and context metrics
presidio-analyzer	2.2.364	Identifier detection on egress
faster-whisper	1.2.1	Local speech-to-text for the Pro tier

Versions move. The team pins after its own check, writes the date next to the pin, and treats any number in a tutorial older than the version it runs as unverified.

CONTEXT

Three depth tiers on one scenario: tiers set depth, never access

CORE · ABOUT 38 HOURS

Ship the two patient workflows

- W1 end to end with checkpointer and interrupt.
- W2 ingest and ask with the scope guard and citations.
- D1 to D5 filed; audit event on every record read.
- Thirty eval cases, red-flag recall at 100%.
- One-page memo and a six-minute demo.

Done when a patient books and asks about a record in the demo, and the red-team PDF fails to leak.

PRO · ABOUT 58 HOURS

Add the doctor, and run it like production

- Everything in Core.
- W3 scribe with the doctor interrupt and edit history.
- Async ingest on a queue; model routing with a fallback.
- D6 and D7; a cost sheet from real traces.
- Eighty eval cases with a calibrated judge; tracing on a dev run.

Done when a doctor signs a stock-flagged note in the demo and the cost per consultation is shown with its formula.

ARCHITECT · ABOUT 80 HOURS

Prove it at the client's scale

- Everything in Pro.
- An MCP tool server exposing scheduling and stock to any agent.
- Parallel ingest with the Send API under a bounded pool.
- A load test at 1,400 requests an hour, p95s on screen.
- The hybrid deployment ADR with the egress boundary implemented.
- One roadmap extension from page 52.

Done when the load test passes under the stated p95s and the ADR survives the steering review.

DESIGN

Week one: discovery closes, the data estate stands, W1 goes live

DAYS 1 TO 2

Discovery

D1 automation ladder, D2 questions posted, D3 assumptions register, the data inventory with exact counts and licences, and the repository scaffolded.

Gate: discovery closes on the page 10 checklist.

DAYS 3 TO 4

Data estate

Clean loads of all five sources into Postgres, the entity model migrated, the generated layer at version one, D4 contracts and D5 sizing filed.

Gate: every transformation has its passing check.

DAY 5

Architecture and evals

Architecture pack at version one: context, container, the W1 graph and ADRs 1 to 3; the eval set of thirty cases written before any graph code.

Gate: evals exist, graph code does not yet.

DAYS 6 TO 7

W1 live

W1 end to end with the checkpointer and the interrupt, first traces captured, red-flag recall measured, and a week-one status note sent to the client thread.

Gate: red-flag recall at 100% on the set.

DESIGN

Week two: records, the change request, assurance, and the steering review

DAYS 8 TO 9

W2 and the change

Ingest and ask graphs live with the scope guard and citations. The sealed change request arrives on day 8.

Two hours to re-plan; an ADR records what moved.

DAYS 10 TO 11

Depth days

Pro pods ship W3 with the doctor interrupt. Architect pods add the MCP server and parallel ingest. Core pods harden W1 and W2.

Every pod: traces on every run.

DAY 12

Assurance

Full eval run, the cost sheet from traces, D6 and D7 filed, and the observability view with p95s and red-flag rate.

Numbers, not adjectives, from here on.

DAY 13

Memo and demo

One-page executive memo, the six-minute demo recording, repository hygiene and the AI-collaboration log complete.

The memo opens with the business result.

DAY 14

Submission

Pull request to the cohort repository, non-code artefacts in the Discussions thread, and the steering review booked within 48 hours.

Late means the tier below.

ASSURANCE

The change request: sealed until day 8, and scored on how the team absorbs it

Held by the delivery office

Real clients change their minds in week two. One scoped change lands in the client thread on day 8, in the client's own voice, and it is not in this pack.

What is known now: it touches one of data, regulation or integration; it is achievable inside the remaining days by trading scope; and it is the same for every pod in a cohort.

THE TWO-HOUR RE-PLAN, IN ORDER

10 min	Read it twice and restate it to the client in one sentence; the client confirms before anything else happens.
30 min	Trace the impact across scope, data model, architecture and the eval set, and write it down as a list, not a feeling.
30 min	Lay out three options with their costs: absorb it, trade it against something in scope, or defer it to the roadmap.
20 min	Decide in an ADR and tell the client what moves, the same day, in the thread.
30 min	Re-plan the board for days 9 to 14 and add the change's cases to the eval set before touching code.

The steering review scores the ADR, what was cut and why, and whether the client heard it the same day; it does not score whether the change was fully built.

ASSURANCE

Pre-mortem: six ways this engagement fails, written before the first node

Failure	Early signal	Detection the team builds	Response
A patient sees another patient's chunk	A citation whose doc_id is not in the caller's record	Foreign-id check on every answer; the red-team PDF in the eval set	Block the answer, write an incident, review the retrieval filter
A medicine is invented in a draft	A molecule in P that is absent from the transcript	Transcript-grounding check on every medicine line	Replace with the "doctor to specify" bracket, log the case
A red flag is missed	An emergency presentation routed to a routine slot	Red-flag recall measured on every eval run, target 100%	Stop the release; add the case; widen the rule set
A slot is double-booked	Two appointment rows with one slot_id	A unique constraint and an idempotency test under retries	Front desk override; a client apology that names the fix
Cost runs away	Tokens per triage turn climbing week on week	A cost meter per run with a daily budget alert	Route the step to a small model; cap turns per session
Doctors ignore the drafts	Review time rising, edits touching every field	Edit distance and review time on the dashboard	Shadow five consultations; fix the template before the model

The pre-mortem is filed with the architecture pack on day 5 and reread on day 12; a failure that happened without its signal firing is a finding for the memo, not a secret.

ASSURANCE

Evaluation sets: one golden set per workflow, written before the graph

Workflow	Each case holds	Core / Pro	Primary metrics	Hard cases that must be in
W1 triage	An intake message, the expected specialty and the red-flag label	30 / 80	Specialty accuracy; red-flag recall at 100%	Mixed Hindi-English phrasing; two symptoms pointing at two specialties
W2 ask	A question, the patient's record and the expected answer span	30 / 80	Faithfulness; citation precision; scope leaks at zero	A question about another patient; a question outside the record; the red-team PDF
W3 scribe	A transcript and the expected SOAP fields and medicine list	Pro only / 40	Field-level F1; medicine precision; invented molecules at zero	Overlapping speech; brand versus generic names; a class named without a molecule
Platform	A load profile and the p95 targets	Pro / Architect	p95 per graph; error rate under load	Retry storms; the LLM API returning errors for one

ONE CASE, AS STORED

```
{ "id": "w1-017",
"input": "dull ache left arm,
breathless since morning, age 62,
diabetic",
"expect": { "red_flag": true,
"route": "emergency_desk" },
"must_not": ["specialty", "slot"],
"source": "pack page 14, case A",
"added_by": "pod lead", "version":
1 }
```

Every case names its source and its author, so a failing case can be argued about and a passing one trusted.

ASSURANCE

Five ways to judge an answer, from cheapest to most trusted

Deterministic checks

Schema valid, every citation resolves to a retrieved chunk, no foreign doc_id, no molecule absent from the transcript. Free, fast, run on every call.

Reference-based

Exact match for specialty and red-flag labels; field-level F1 for SOAP sections; precision and recall for medicine lists against the golden set.

Model as judge

Faithfulness and context precision with ragas 0.4.x, calibrated against twenty human-graded samples first; never the only gate on a release.

Human review

A doctor in role spot-checks ten percent of drafts; disagreements with the judge recalibrate it. Review time itself is a metric.

Red-team set

Injected PDFs, prompts that ask for another patient, requests for doses and diagnoses. Zero tolerance: one leak fails the release.

No library ships a scope-leak test for this data model; the team writes it as a deterministic check and it is the first thing the client's architect asks to see.

ASSURANCE

Observability: the operations screen the client sees, and what feeds it

p95 triage turn	p95 record answer	Red-flag recall	Scope leaks	Review per note	Cost per run
under 4 s	**under 8 s**	**100%**	**zero**	**under 90 s**	**computed**
from traces	from traces	from eval runs	from red-team set	from the queue	from the cost meter

Traces

Every run carries a trace id with one span per node, tool call and model call, including token counts; Langfuse 5.x or LangSmith hold them, and the trace link is printed in every audit event.

Metrics

p95 per graph, cost per run, interrupt wait time, red-flag rate and the edit distance between draft and signed note; each tile above is one query over traces or eval runs.

Review queue

Human-in-the-loop items with an owner and a service level: who, what, how long waiting; everything reviewed feeds the eval set, which is how the system learns without retraining.

The tiles show targets, not results: the team's dashboard replaces each target with the measured figure from its own synthetic runs, and a tile with no source is removed.

REVIEW

What the team hands over, where it lives, and how it is submitted

REPOSITORY LAYOUT

```
docs/ rfp-response.md · prd-lite.md · adr/ · diagrams/
docs/ task-cards.md · assumptions.md · data-contracts/
data/ inventory.md · transforms/ · synthetic/prompts/
services/ agent/ · scheduling/ · records/ · scribe/
evals/ sets/ · runs/ · report.md
infra/ docker-compose.yml · deployment-adr.md
memo/ executive-memo.md
AI-COLLABORATION.md
```

Deliverable	Accepted when
Working repository	One command starts the stack; the README names the tier reached.
Architecture pack	Context, container, entity model, three graphs, two sequences, all corrected from this pack.
Discovery deliverables D1 to D7	Each filed at its path with the acceptance line on its page met.
Evaluation report	Every metric on page 16 with the measured figure, or the reason it is missing.
Executive memo, one page	Opens with the business result; written as an interview case study.
Demo recording, six minutes	Shows a booking, a cited answer and, for Pro, a signed note.
AI-collaboration log	One entry per pull request; pass or fail, not scored.

Code is submitted as a pull request to the cohort repository; the memo, the pack and the report go to the Discussions thread; the steering review is booked within 48 hours of the pull request.

REVIEW

Scoring: four weighted factors that mirror the client's own evaluation criteria

20	**Problem understanding and solution design**	**Full marks:** the ladder, questions and assumptions trace to design decisions, and the architecture pack reads as one argument. **Loses marks:** diagrams copied from this pack uncorrected; scope that serves no persona.
40	**Technical implementation and functionality**	**Full marks:** graphs run end to end with checkpointer, interrupt and tools; the scope guard and red-flag gate hold under test; the code is readable. **Loses marks:** prompts doing the job of code; secrets or identified data in the repo; a demo that works once.
25	**Evaluation, results and responsible AI**	**Full marks:** measured figures against page 16, the red-team set passing, cost and threat deliverables filed. **Loses marks:** metrics quoted without a run; a judge with no calibration; no audit trail.
15	**Communication and presentation**	**Full marks:** the memo opens with the business result, the demo runs six minutes, and review answers are specific. **Loses marks:** a memo that narrates effort; a demo of settings screens.
Gate	**AI-collaboration log, pass or fail:** one entry per pull request saying what the agent proposed, what the team rejected and why. A repository without it is not scored, whatever the code does. Scores sum to 100 and are shared with the pod after the review.	

REVIEW

The steering review: twenty minutes, and the eight questions the client's architect asks

Format

Six minutes: the demo recording plays; nobody narrates over it.

Five minutes: a technical defence on one node, one decision and one failure, chosen by the reviewer from the repository, not by the pod.

Nine minutes: the ADRs and the change request, in the client's voice.

Held within 48 hours of the pull request. Any pod member may be asked any question; the pod lead may not answer for them.

The architect asks	A senior answer contains
Why a graph and not a chain of calls?	Explicit state, interrupts, checkpoints and replay; the moment a chain stopped being enough.
How can a patient never see another's data?	The filter in the SQL at retrieval, not in the prompt; the red-team test that proves it.
What happens when the model API is down?	Timeouts, a circuit breaker and degradation to the human queue; the trace that shows it.
How do you know a cited answer is right?	Deterministic citation checks, a calibrated judge, a doctor sampling ten percent.
On-premise or cloud, and why?	The residency constraint decides; hybrid with de-identified egress; the trigger to switch.
What did the change request cost you?	The ADR, what was cut, and the message the client received the same day.
What does a consultation cost to run?	The worksheet, the price-list date, and which line model routing moved.
Where did your coding agent mislead you?	A specific log entry, what was rejected, and the test that would have caught it.

REVIEW

The interview angle: what each stage of this engagement lets an FDE say in a hiring room

Stage	The question hiring managers ask	What a senior answer contains, from this engagement
Discovery	“Tell me about a time you pushed back on a client requirement.”	The automation ladder: which task the client wanted automated, why it landed on the human rung, and how the client took it.
Data	“How did you handle sensitive data in an LLM system?”	Four sensitivity classes, the scrubber on egress, the token map on-premise, and the red-team PDF that proved the filter.
Architecture	“When would you use LangGraph over a simpler orchestration?”	Interrupt and resume for the doctor’s signature, checkpoints that make the API stateless, and the moment a chain stopped being enough.
Platform	“How do you decide between cloud and on-premise for regulated data?”	The scored matrix, the residency constraint that decided it, and the trigger that would reverse the decision.
Delivery	“Describe a scope change mid-project and what you did.”	The day-8 change request, the two-hour re-plan, the ADR and the same-day message to the client.
Assurance	“How do you evaluate a RAG system beyond vibes?”	Golden sets written before code, faithfulness and citation precision, a calibrated judge, zero-tolerance scope leaks.
Economics	“What does your system cost to run, per transaction?”	The worksheet, the measured token counts from traces, and which line model routing moved.
Ways of working	“How do you use AI coding tools responsibly?”	Spec and evals first, every diff read, the AI-collaboration log, and one concrete time the agent was wrong.

The one-page memo is written as this table’s cover story: a hiring manager should finish it knowing what the client got, what the team decided, and what the team would do differently.

Roadmap: five extensions the client has asked about, and what each proves

PRO

Nursing notes summarised

Inpatient progress notes rolled into a shift summary for the ward round, with every line cited to its note.

Proves long-context summarisation with citations.

PRO

Real-time pharmacy feed

The nightly CSV adapter is swapped for the promised API behind the same interface, with staleness shown on every flag.

Proves the adapter boundary held.

ARCHITECT

MCP tool server

Scheduling and stock exposed as tools over the Model Context Protocol (mcp 2.3.x), so any approved agent in the hospital can book or check stock under the same audit.

Proves tools as a product, not a private function.

ARCHITECT

Claims and coding assist

ICD-10 code suggestions on a signed note for a human coder to accept or change, with a pre-authorisation packet assembled from the record.

Proves human-in-the-loop on a financial act.

ARCHITECT

Supervisor with specialists

A routing agent that hands intake, records and scribe work to specialist graphs and fans out with the Send API, under one trace.

Proves multi-agent control without losing the audit.

None of the five is a commitment to the client; each is an option priced in the memo. An Architect pod builds one and writes the other four as one-paragraph proposals.

CONTEXT

Annex: the three templates every deliverable is written on

Task card, for D1

```
task: one verb phrase
rule_known: yes | partly | no
input_bounded: yes | no
cost_of_error: who notices, what it costs
sign_off: role or none
rung: code | llm | agentic | human
why: two sentences
eval_cases: ids
```

Decision record, ADR

```
title: the decision in one line
context: the constraint that forced it
options: three, with costs
decision: one, with the trigger to revisit
consequences: what gets harder
versions_pinned: package == x.y.z
checked_on: date of the check
status: proposed | accepted | superseded
```

Data contract, for D4

```
table: name
owner: a role, never a person
schema_version: n, change rule
freshness: max age before consumers must know
sensitivity: direct | quasi | clinical | operational
checks: nulls, ranges, uniqueness, thresholds
consumers: who breaks when this changes
source: dataset and licence
```

Templates are filled as plain text in the repository so they diff in pull requests; a template with an empty field is filed as incomplete, never as blank.

Glossary: sixteen terms the pack uses the way the client's teams use them

Term	Meaning here
FDE	A forward-deployed engineer: builds inside the client's problem, owns the outcome, not only the code.
RFP	Request for proposal: the client's statement of what it wants to buy and how it will judge bids.
PRD-lite	A short requirements document: personas, journeys, epics with testable acceptance criteria.
Interrupt	A LangGraph pause inside a run that waits for a human input and resumes from the checkpoint.
Checkpointer	The store that saves graph state after each step, so a run can pause, resume and be replayed.
RAG	Retrieval-augmented generation: the model answers from retrieved documents and cites them.
Scope leak	Any answer that uses data the caller is not allowed to see; the one metric held at zero.
Identified data	Health data that can be tied to a person; it never leaves India or the hospital boundary.

Term	Meaning here
DPDP Act	India's Digital Personal Data Protection Act, 2023; its Rules were notified in November 2025 with staged commencement.
ABDM, ABHA	India's digital health mission and the patient health id it issues; linking it is an extension here.
FHIR R4	The HL7 standard for exchanging health records; the client's hospital system exposes it read-only.
MCP	Model Context Protocol: a standard for exposing tools and data to agents; the Architect extension.
ADR	Architecture decision record: one decision, its options, its consequences and its revisit trigger.
SOAP note	A clinical note in four parts: subjective, objective, assessment, plan.
pgvector	The Postgres extension that stores embeddings and runs similarity search next to the rows.
p95	The latency that 95 percent of requests beat; the figure operations teams quote.

CONTEXT

Reading list: every source this pack rests on, with the date it was checked

Source	Where	What it settles here
LangGraph 1.2.16 and its Postgres checkpointer 3.2.0	pypi.org/project/langgraph · pypi.org/project/langgraph-checkpoint-postgres	The version pins and release dates behind every graph page; checked 21 Oct 2026.
Model Context Protocol SDK 2.3.0	pypi.org/project/mcp · modelcontextprotocol.io	The Architect-tier tool server.
langfuse 5.0.3, ragas 0.4.3, presidio-analyzer 2.2.364, faster-whisper 1.2.1	pypi.org/project/‹name›	Tracing, evaluation, identifier detection and local speech-to-text; checked 21 Oct 2026.
DPDP Rules 2025, phased commencement	azbpartners.com, “phased rollout and key compliance milestones”	Notified 13 Nov 2025; consent-manager rules from 12 Nov 2026; substantive obligations from 12 May 2027.
HL7 FHIR R4	hl7.org/fhir/R4	The read-only integration contract with the hospital system.
Ayushman Bharat Digital Mission	abdm.gov.in	The ABHA identifier and ABDM readiness named in the mandate.
The five public datasets	kaggle.com/datasets/‹owner›/‹slug› as on page 17 · openml.org dataset 43439	Existence verified 21 Oct 2026; row counts and licences recorded by the team in its inventory.

Anything not on this list is the delivery office’s own construction and is labelled so where it appears: the automation ladder, the house rules, the two-hour re-plan and the fictional client.

FDE ACADEMY · DELIVERY OFFICE

Build what a client would pay for, then explain it like a senior.

BEFORE THE REPOSITORY

D1 to D3 filed and discovery closed on the page 10 checklist.

BEFORE CODE

The eval set, the architecture pack and the pre-mortem, all on day 5.

BEFORE THE REVIEW

The memo that opens with the business result, and one honest line about where the agent misled you.